
WIND TURBINE GENERATOR POWER CURVE VERIFICATION TERRAIN ASSESSMENT REPORT

Pestera2 Wind Farm Project
Romania

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TABLE OF CONTENTS

1. Executive Summary
2. Project Description & Input Data
 - 2.1 Wind Farm Layout
 - 2.2 External Wind Farms
 - 2.3 Wind Climate Data
 - 2.4 Terrain Map Information
3. Measurement Sector Analysis
 - 3.1 Proposed Mast Locations
 - 3.2 Terrain Assessment Results
 - 3.3 Terrain Classification Summary
4. Freestream Sector Analysis
5. PCV Site Selection & Optimization
 - 5.1 Mast-WTG Pairing Scores
 - 5.2 1-Mast-2-WTGs Configuration
 - 5.3 Mast Location Proposals
6. Final Measurement Sectors
7. Recommendations & Conclusions

1. EXECUTIVE SUMMARY

This report presents the results of the IEC 61400-12-1 terrain assessment and Power Curve Verification (PCV) site suitability analysis for the Pestera2 Wind Farm project located in Romania. The analysis covers 55 internal wind turbine generators (WTGs) of type N163-7.0MW and considers the wake influence from 43 external V90-3MW turbines in the vicinity. The assessment was performed using the WTG PCV Tool v1.0, following the requirements of IEC 61400-12-1:2017 (Edition 2).

Key Findings:

Parameter	Value	Assessment
Internal WTGs Analyzed	55	N163-7.0MW (D=163m, HH=138m)
External WTGs Considered	43	V90-3MW (D=90m, HH=90m)
Proposed Met Masts	3	Lattice type, 138m height
Total Mast-WTG Assessments	165	36 sectors each
Average Valid Sectors	2.1 of 36	REVIEW
Average Freestream Sectors	2.1 of 36	MODERATE
Best PCV Configuration Score	46.9/100	CONDITIONAL
Best Combined Valid Sectors	10	for best configuration
Mean Wind Speed (from data)	7.2 m/s	Jan-Feb 2004
Terrain Elevation Range	45 - 145 m	from .map file

Key Recommendations:

- Recommended mast: Proposed Mast 2 - Dual WTG Testing with overall suitability score 46.9/100.
- Target WTG: T56 (N163-7.0MW) (46.9/100 score).
- Warning: Valid sector coverage is below 50%. Consider additional mast placement or sector filtering adjustments.
- Best individual pairing: Mast "Proposed Mast 2 - Dual WTG Testing" → WTG "T56 (N163-7.0MW)" (7.8D distance, 46.9/100 score).

2. PROJECT DESCRIPTION & INPUT DATA

The Pesteră2 Wind Farm is located in Romania and consists of 55 Nordex N163-7.0MW wind turbine generators, each with a rotor diameter of 163m and hub height of 138m. The wind farm layout was provided in Romanian Stereo70 projection (EPSG:7755) and has been converted to WGS84 for this analysis. The project area also has 43 external Vestas V90-3MW turbines (D=90m, HH=90m) in the surrounding area that may influence the freestream conditions at the proposed measurement mast locations. The external turbine layout was provided to enable a comprehensive wake impact assessment for accurate freestream sector determination, which is critical for IEC 61400-12-1 compliant power curve verification testing.

2.1 Wind Farm Layout (Internal WTGs)

The internal wind farm spans approximately 8.2 km (N-S) by 6.4 km (E-W), covering a total area of approximately 52.9 km². All 55 turbines are of identical type N163-7.0MW, ensuring uniform wake characteristics across the farm. The WTG coordinates were provided in the Romanian Stereo70 national projection system (EPSG:7755) and have been transformed to WGS84 geographic coordinates for this analysis. Coordinate accuracy is within +/-3m as specified in the source data.

ID	Model	Lat	Lon	RD (m)	HH (m)
T1	T1	44.3323	28.2222	163.0	138.0
T2	T2	44.3401	28.2213	163.0	138.0
T4	T4	44.2957	28.1764	163.0	138.0
T5	T5	44.3513	28.2144	163.0	138.0
T6	T6	44.3381	28.2030	163.0	138.0
T7	T7	44.3362	28.2178	163.0	138.0
T8	T8	44.3437	28.2174	163.0	138.0
T9	T9	44.3462	28.2133	163.0	138.0
T10	T10	44.3275	28.2177	163.0	138.0
T12	T12	44.3385	28.2123	163.0	138.0
T13	T13	44.3328	28.2088	163.0	138.0
T14	T14	44.3501	28.2075	163.0	138.0
T15	T15	44.3252	28.2111	163.0	138.0
T17	T17	44.3274	28.1754	163.0	138.0
T18	T18	44.3441	28.2064	163.0	138.0
...	and 40 more				

2.2 External Wind Farms

The external wind farm comprises 43 Vestas V90-3MW turbines with a rotor diameter of 90m and hub height of 90m. These turbines are located to the southwest and east of the main wind farm and have been included in the analysis to assess their wake impact on freestream conditions at the proposed met mast locations. The external turbines represent operational wind farms in the vicinity that could affect the measured wind speed and direction at the measurement site, particularly during prevailing wind directions from the southwest sector.

2.3 Wind Climate Data

Wind climate data was provided as hourly time series from January to February 2004 (total 8753 valid records after filtering calm periods below 0.5 m/s). The mean wind speed during this period was 7.2 m/s with a maximum recorded speed of 19.8 m/s. The predominant wind direction is from 275 degrees (variable). This wind climate information, while based on a limited measurement period, provides valuable context for understanding the expected wind conditions during the PCV measurement campaign. A full wind resource assessment with 12+ months of data is recommended for final campaign planning.

Direction	Freq (%)	Avg WS (m/s)	Max WS (m/s)
275 (W)	4.4	8.3	17.1
295 (WNW)	4.4	7.1	16.1
305 (NW)	4.1	7.0	15.3
285 (WNW)	4.1	7.4	18.8
315 (NW)	4.0	6.8	17.1
265 (W)	3.9	8.1	17.5
335 (NNW)	3.5	6.9	19.1
345 (NNW)	3.5	6.9	19.8
185 (S)	3.5	8.1	14.1
325 (NW)	3.3	6.9	18.3
175 (S)	3.3	8.2	14.7
195 (SSW)	3.1	8.6	17.4

2.4 Terrain Map Information

The terrain data was provided as a WASP .map file (MAPFILES_240102_Pestera2_Rev 0_0.map) in the Romanian Stereo70 coordinate system. The file contains 2481 elevation contours with a total of 171869 coordinate points. Elevation values range from 45m to 145m with a mean of 102m. The contours are at 45m intervals, representing the topographic variation across the site. The terrain is characterized by moderate slopes typical of Romanian wind farm sites, with the majority of the area classified as simple to moderately complex terrain per IEC 61400-12-1 criteria.

3. MEASUREMENT SECTOR ANALYSIS

The measurement sector analysis is performed according to IEC 61400-12-1:2017, which requires assessment of terrain slope, roughness conditions, and wake influence for each of 36 directional sectors (10-degree width each). A sector is considered valid for power curve verification if it meets all of the following criteria: maximum terrain slope does not exceed the threshold for the applicable terrain classification (Class A: slope ≤ 10 degrees for simple terrain, Class B: slope ≤ 17 degrees for complex terrain), no significant roughness changes within the assessment radius, and the sector is not significantly affected by wakes from upstream turbines. The analysis was performed for each proposed mast location against all target WTGs within the wind farm.

3.1 Proposed Mast Locations

3 met mast locations have been proposed for the PCV measurement campaign, strategically positioned to maximize valid sector coverage while maintaining adequate separation from all turbines. Each mast is designed as a 138m lattice tower with measurement levels at 40m, 60m, 80m, 100m, 120m, and 138m heights, matching the hub height of the target N163-7.0MW turbines.

Mast ID	Strategy	Lat	Lon	Min WTG Dist (m)
MAST-PCV-01	4D north of central WTG	44.3259	28.1982	651
MAST-PCV-02	Perpendicular to T8-T9 line at 4D	44.3500	28.2195	435
MAST-PCV-03	Upstream of farm in prevailing wind direction	44.3191	28.1775	156

3.2 Terrain Assessment Results

Mast	Target WTG	Distance (m)	Distance (D)	Valid Sectors	Freestream	Terrain Class	IEC Compliant
Proposed Mast 1 - C811a(N163-7.0MW)	C811a(N163-7.0MW)	2035	12.5	0	0	B	NO
Proposed Mast 1 - C812a(N163-7.0MW)	C812a(N163-7.0MW)	2427	14.9	0	0	B	NO
Proposed Mast 1 - C814a(N163-7.0MW)	C814a(N163-7.0MW)	3782	23.2	0	0	B	NO
Proposed Mast 1 - C815a(N163-7.0MW)	C815a(N163-7.0MW)	3101	19.0	0	0	B	NO
Proposed Mast 1 - C816a(N163-7.0MW)	C816a(N163-7.0MW)	1413	8.7	0	0	B	NO
Proposed Mast 1 - C817a(N163-7.0MW)	C817a(N163-7.0MW)	1935	11.9	0	0	B	NO
Proposed Mast 1 - C818a(N163-7.0MW)	C818a(N163-7.0MW)	2501	15.3	0	0	B	NO
Proposed Mast 1 - C819a(N163-7.0MW)	C819a(N163-7.0MW)	2552	15.7	0	0	B	NO
Proposed Mast 1 - C8110a(N163-7.0MW)	C8110a(N163-7.0MW)	1563	9.6	0	0	B	NO
Proposed Mast 1 - C8112a(N163-7.0MW)	C8112a(N163-7.0MW)	1795	11.0	0	0	B	NO
Proposed Mast 1 - C8113a(N163-7.0MW)	C8113a(N163-7.0MW)	1144	7.0	0	0	B	NO
Proposed Mast 1 - C8114a(N163-7.0MW)	C8114a(N163-7.0MW)	2788	17.1	0	0	B	NO
Proposed Mast 1 - C8115a(N163-7.0MW)	C8115a(N163-7.0MW)	1032	6.3	0	0	B	NO
Proposed Mast 1 - C8117a(N163-7.0MW)	C8117a(N163-7.0MW)	1817	11.1	0	0	B	NO
Proposed Mast 1 - C8118a(N163-7.0MW)	C8118a(N163-7.0MW)	2127	13.0	0	0	B	NO
Proposed Mast 1 - C8119a(N163-7.0MW)	C8119a(N163-7.0MW)	754	4.6	0	0	B	NO
Proposed Mast 1 - C8120a(N163-7.0MW)	C8120a(N163-7.0MW)	1939	11.9	0	0	B	NO

Proposed Mast 1 - Central North (N163-7.0MW)	2111	1898	11.6	0	0	B	NO
Proposed Mast 1 - Central North (N163-7.0MW)	2221	2795	17.1	0	0	B	NO
Proposed Mast 1 - Central North (N163-7.0MW)	2231	2423	14.9	0	0	B	NO
...	and 145 more						

3.3 Terrain Classification Summary

Out of 165 total mast-WTG pair assessments, the terrain classification distribution is as follows: Class A (simple terrain, slope <= 10 degrees): 110 assessments (66.7%), Class B (complex terrain, slope <= 17 degrees): 55 assessments (33.3%), and Class S (special terrain, slope > 17 degrees): 0 assessments (0.0%). Overall, 3 of 165 assessments (1.8%) meet the IEC 61400-12-1 compliance criteria for terrain effects. The majority of the site falls within the simple terrain classification, indicating favorable conditions for power curve verification testing with minimal terrain-induced flow distortion.

Sector-by-Sector Analysis (Representative Pair):

Mast: Proposed Mast 1 - Central North | Target WTG: T1 (N163-7.0MW) | Distance: 2035m (12.5D)

Sector	Dir Range	Max Slope	Slope (deg)	Class	Valid	Freestream	Reason
0	-5-5	21.2%	12.0	B	YES	NO	Maximum slope 12.0° exceeds Cl...
10	5-15	21.2%	12.0	B	YES	NO	Maximum slope 12.0° exceeds Cl...
20	15-25	21.2%	12.0	B	YES	NO	Maximum slope 12.0° exceeds Cl...
30	25-35	21.2%	12.0	B	YES	NO	Maximum slope 12.0° exceeds Cl...
40	35-45	21.2%	12.0	B	YES	NO	Maximum slope 12.0° exceeds Cl...
50	45-55	21.0%	11.9	B	YES	NO	Maximum slope 11.9° exceeds Cl...
60	55-65	21.0%	11.9	B	YES	NO	Maximum slope 11.9° exceeds Cl...
70	65-75	21.0%	11.9	B	YES	NO	Maximum slope 11.9° exceeds Cl...
80	75-85	20.8%	11.7	B	YES	NO	Maximum slope 11.7° exceeds Cl...
90	85-95	20.8%	11.7	B	YES	NO	Maximum slope 11.7° exceeds Cl...
100	95-105	20.8%	11.7	B	YES	NO	Maximum slope 11.7° exceeds Cl...
110	105-115	20.8%	11.7	B	YES	NO	Maximum slope 11.7° exceeds Cl...
120	115-125	20.6%	11.6	B	YES	NO	Maximum slope 11.6° exceeds Cl...
130	125-135	20.6%	11.6	B	YES	NO	Maximum slope 11.6° exceeds Cl...
140	135-145	20.6%	11.6	B	YES	NO	Maximum slope 11.6° exceeds Cl...
150	145-155	20.6%	11.6	B	YES	NO	Maximum slope 11.6° exceeds Cl...
160	155-165	20.6%	11.6	B	YES	NO	Maximum slope 11.6° exceeds Cl...
170	165-175	20.6%	11.6	B	YES	NO	Maximum slope 11.6° exceeds Cl...
180	175-185	20.6%	11.6	B	YES	NO	Maximum slope 11.6° exceeds Cl...
190	185-195	20.6%	11.6	B	YES	NO	Maximum slope 11.6° exceeds Cl...
200	195-205	20.6%	11.6	B	YES	NO	Maximum slope 11.6° exceeds Cl...
210	205-215	20.6%	11.6	B	YES	NO	Maximum slope 11.6° exceeds Cl...
220	215-225	20.8%	11.7	B	YES	NO	Maximum slope 11.7° exceeds Cl...
230	225-235	20.8%	11.7	B	YES	NO	Maximum slope 11.7° exceeds Cl...
240	235-245	20.8%	11.7	B	YES	NO	Maximum slope 11.7° exceeds Cl...
250	245-255	21.0%	11.9	B	YES	NO	Maximum slope 11.9° exceeds Cl...

260	255-265	21.0%	11.9	B	YES	NO	Maximum slope 11.9° exceeds Cl...
270	265-275	21.0%	11.9	B	YES	NO	Maximum slope 11.9° exceeds Cl...
280	275-285	21.0%	11.9	B	YES	NO	Maximum slope 11.9° exceeds Cl...
290	285-295	21.2%	12.0	B	YES	NO	Maximum slope 12.0° exceeds Cl...
300	295-305	21.2%	12.0	B	YES	NO	Maximum slope 12.0° exceeds Cl...
310	305-315	21.2%	12.0	B	YES	NO	Maximum slope 12.0° exceeds Cl...
320	315-325	21.2%	12.0	B	YES	NO	Maximum slope 12.0° exceeds Cl...
330	325-335	21.2%	12.0	B	YES	NO	Maximum slope 12.0° exceeds Cl...
340	335-345	21.2%	12.0	B	YES	NO	Maximum slope 12.0° exceeds Cl...
350	345-355	21.2%	12.0	B	YES	NO	Maximum slope 12.0° exceeds Cl...

4. FREESTREAM SECTOR ANALYSIS

The freestream sector analysis identifies measurement directions that are not significantly influenced by wake effects from upstream wind turbines. This analysis considers both internal WTGs within the project wind farm and external WTGs from neighboring wind farms. A sector is classified as freestream if no upstream turbine falls within the wake angular threshold (30 degrees) and wake distance threshold (20 rotor diameters) for that direction. The Jensen wake model with a 7.5-degree expansion angle is used to estimate wake width at the mast location, and sectors with combined wake impact rated as 'low' or higher are excluded from the freestream sector set. This is critical for ensuring that the wind speed measurements used for power curve verification are representative of undisturbed (freestream) conditions.

Mast	Freestream Sectors	Wake-Affected	Freestream %	External WTGs Imp
Proposed Mast 1	Central North	36	0.0%	0
Proposed Mast 2	Downstream, 340, 350	30	0.0%	0
Proposed Mast 3	Upstream Southwest	36	0.0%	0

5. PCV SITE SELECTION & OPTIMIZATION

The PCV (Power Curve Verification) site selection process evaluates all possible mast-WTG pairings based on five weighted criteria: distance optimality (20%), terrain quality (25%), sector coverage (20%), freestream quality (25%), and slope compliance (10%). The optimization also considers configurations where a single met mast can test multiple WTGs (1-mast-2-WTGs), which is cost-effective and practically feasible when the mast location provides adequate valid sector coverage for both target turbines simultaneously. The following sections present the pairing scores, optimal configurations, and proposed mast locations.

5.1 Top Mast-WTG Pairing Scores

Mast	WTG	Score	Valid Sec	Free Sec	Distance (D)	Recommended
Proposed Mast 2T56 (N163-7.0MW)	T56 (N163-7.0MW)	46.9	10	10	7.8	-
Proposed Mast 2T54 (N163-7.0MW)	T54 (N163-7.0MW)	44.9	9	9	8.0	-
Proposed Mast 2T50 (N163-7.0MW)	T50 (N163-7.0MW)	44.9	9	9	7.1	-
Proposed Mast 2T50 (N163-7.0MW)	T50 (N163-7.0MW)	41.0	7	7	2.7	-
Proposed Mast 2T20 (N163-7.0MW)	T20 (N163-7.0MW)	39.1	6	6	6.8	-
Proposed Mast 2T70 (N163-7.0MW)	T70 (N163-7.0MW)	39.1	6	6	9.4	-
Proposed Mast 2T80 (N163-7.0MW)	T80 (N163-7.0MW)	39.1	6	6	4.4	-
Proposed Mast 2T90 (N163-7.0MW)	T90 (N163-7.0MW)	39.1	6	6	4.0	-
Proposed Mast 2T12 (N163-7.0MW)	T12 (N163-7.0MW)	39.1	6	6	8.6	-
Proposed Mast 2T12 (N163-7.0MW)	T12 (N163-7.0MW)	39.1	6	6	5.9	-

5.2 Mast Configurations (1-Mast Testing Multiple WTGs)

Configuration: Proposed Mast 2 - Dual WTG Testing - Overall Score: 46.9/100 | Combined Valid Sectors: 10 | Combined Freestream: 10 | Recommended: NO

Target WTG	Score	Valid Sectors
T56 (N163-7.0MW)	46.9	10

Note: Single WTG PCV configuration

Configuration: Proposed Mast 2 - Dual WTG Testing - Overall Score: 46.1/100 | Combined Valid Sectors: 13 | Combined Freestream: 13 | Recommended: NO

Target WTG	Score	Valid Sectors
T56 (N163-7.0MW)	46.9	10
T54 (N163-7.0MW)	44.9	9

Note: 1 mast (Proposed Mast 2 - Dual WTG Testing) testing 2 WTGs: T56 (N163-7.0MW) & T54 (N163-7.0MW)

Note: Combined valid sector coverage: 36.1%

Configuration: Proposed Mast 3 - Upstream Southwest - Overall Score: 25.2/100 | Combined Valid Sectors: 0 | Combined Freestream: 0 | Recommended: NO

Target WTG	Score	Valid Sectors
T17 (N163-7.0MW)	25.2	0

Note: Single WTG PCV configuration

5.3 Additional Mast Location Proposals

Proposal ID	Target WTGs	Quality Score	Expected Valid	Expected Freestream
candidate_T57_6D_13557		82.0	28	28
candidate_T57_6D_18057		82.0	28	28
candidate_T57_8D_13557		82.0	28	28
candidate_T57_8D_18057		82.0	28	28
candidate_T57_4D_18057		80.0	27	27
candidate_T57_4D_22557		80.0	27	27
candidate_T57_6D_22557		80.0	27	27
candidate_T57_8D_22557		80.0	27	27
candidate_T31_8D_0 T31		78.0	26	26
candidate_T54_8D_0 T54		78.0	26	26

6. FINAL MEASUREMENT SECTORS

The final measurement sectors represent the combination of terrain-valid and freestream sectors for the best PCV configuration. Only sectors that pass both terrain assessment criteria AND are classified as freestream (no significant wake influence) are included in the final set. These sectors define the directional ranges from which wind speed and direction measurements will be used for the power curve verification analysis, ensuring compliance with IEC 61400-12-1 requirements for representative and undisturbed flow conditions.

Mast: Proposed Mast 2 - Dual WTG Testing | Target WTGs: T56 (N163-7.0MW) | Sector Width: 10 deg | Total Coverage: 27.8%

Valid Terrain Sectors: 36 | Freestream Sectors: 10 | Final Combined Sectors: 10

Final Measurement Sectors (degrees): 0, 10, 20, 30, 40, 50, 320, 330, 340, 350

Terrain Class: A | All Criteria Met: YES

- Sector coverage is 27.8%. Consider extending measurement period or adding data sources.

7. RECOMMENDATIONS & CONCLUSIONS

1. The best PCV configuration score of 46.9/100 indicates that site conditions are moderately challenging for PCV testing. Additional mast positions should be evaluated to improve sector coverage and freestream conditions.
2. A minimum 12-month measurement campaign is recommended to capture seasonal wind variability and ensure statistical robustness of the power curve verification results. The measurement equipment should include calibrated anemometers at multiple heights (minimum 3 levels including hub height) and a wind vane.
3. The external wind farm wake influence should be continuously monitored during the measurement campaign, particularly for sectors where the freestream percentage is below 50%. Real-time wake detection algorithms are recommended to supplement the static sector classification performed in this analysis.
4. For the 1-mast-2-WTGs configuration, ensure that the mast location provides adequate fetch and terrain representativity for both target turbines independently. If the combined sector coverage is insufficient, consider installing separate measurement masts for each turbine.
5. Coordinate re-verification of all mast and WTG positions using differential GPS is recommended prior to mast installation, with target accuracy of +/-0.5m to minimize uncertainty in the terrain slope and distance calculations.

Analysis Warnings:

- Mast "Proposed Mast 2 - Dual WTG Testing" to WTG "T4 (N163-7.0MW)" distance (42.6D) is very large. Terrain representativeness may be reduced.
- Mast "Proposed Mast 2 - Dual WTG Testing" to WTG "T50 (N163-7.0MW)" distance (42.0D) is very large. Terrain representativeness may be reduced.
- Mast "Proposed Mast 2 - Dual WTG Testing" to WTG "T51 (N163-7.0MW)" distance (43.5D) is very large. Terrain representativeness may be reduced.
- Mast "Proposed Mast 2 - Dual WTG Testing" to WTG "T52 (N163-7.0MW)" distance (42.2D) is very large. Terrain representativeness may be reduced.
- Mast "Proposed Mast 2 - Dual WTG Testing" to WTG "T55 (N163-7.0MW)" distance (46.9D) is very large. Terrain representativeness may be reduced.
- Mast "Proposed Mast 2 - Dual WTG Testing" to WTG "T57 (N163-7.0MW)" distance (49.1D) is very large. Terrain representativeness may be reduced.
- Mast "Proposed Mast 2 - Dual WTG Testing" to WTG "T60 (N163-7.0MW)" distance (46.4D) is very large. Terrain representativeness may be reduced.
- Mast "Proposed Mast 3 - Upstream Southwest" to WTG "T36 (N163-7.0MW)" distance (1.0D) is below minimum IEC requirement (2D).

END OF REPORT